

A quasi-steady state model for predicting the heating requirements of conventional greenhouses in cold regions

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ABSTRACT

A time-dependent, quasi-steady state thermal model (GREENHEAT) based on the lumped estimation of heat transfer parameters of greenhouses has been developed to predict the hourly heating requirements of conventional greenhouses. The model was designed to predict the hourly heating requirements based on the input of greenhouse indoor environmental control parameters, physical and thermal properties of crops and construction materials, and hourly weather data including temperature, relative humidity, wind speed, and cloud cover. The model includes all of the heat transfer parameters in greenhouses including the heat loss for plant evapotranspiration, and the heat gain from environmental control systems. Results show that the predicted solar radiation data from the solar radiation sub-model are a reasonable fit with the data from the National Solar Radiation Database (NSRDB). Thermal analysis indicates environmental control systems could reduce 13–56% of the total heating requirements over the course of a year in the study greenhouse. During the winter season, the highest amount of greenhouse heat is lost due to conduction and convection, and the heat used for evapotranspiration is dominant in the summer. Finally, the model was validated with actual heating data collected from a commercial greenhouse located in Saskatoon, and the results show that the model satisfactorily predicts the greenhouse heating requirements.

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1. Introduction

Greenhouse production of vegetables can enable people living in cold regions to enjoy fresh healthy food during the winter

season. The harvested area of greenhouse production in Canada has been increasing steadily despite high heating costs [1]. At high northern latitudes, heating of a greenhouse for about eight months of the year is essential to ensure the growth and development of crops grown therein. In Canada, heating accounts for 10–35% of the total greenhouse production costs; the amount of heat necessary depends on the building envelope, the location of the greenhouse, and the kind of crops grown [2]. Different types of thermal models are available that can be used for studying the greenhouse

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Nomenclature

A_c, A_f, A_p	area of cover, floor, and plant, m^2	Q	heat transfer rate, W
A_n, A_t	area of non-transparent and transparent surfaces, m^2	R_a, R_s	aerodynamic resistance and stomatal resistance, $s\ m^{-1}$
CF	cloud cover factor, Octas	Re	Reynold number, dimensionless
C_{pa}	specific heat of air, $J\ kg^{-1}\ K^{-1}$	S	total solar radiation entering the greenhouse, W
E_m	motor efficiency, %	T_c, T_i, T_o	cover temperature, indoor temperature, and outdoor temperature, K
F_c, F_{sk}	cover view factor, and sky view factor, dimensionless	T_s, T_{sk}	underground soil temperature and sky temperature, K
F_p	perimeter heat loss factor, $W\ m^{-1}\ K^{-1}$	T_R	turbidity Factor, dimensionless
F_{hc}, F_a	heat conversion factor, and lighting allowance factor, dimensionless	U_t, U_n	heat transfer coefficient for transparent and non-transparent surfaces, $W\ m^{-1}\ K^{-1}$
F_{um}, F_{ul}	motor load factor, and motor use factor, dimensionless	V	volume of greenhouse, m^3
G_r	Grashof number, dimensionless	v_i, v_o	indoor airspeed, and outdoor airspeed, $m\ s^{-1}$
g	acceleration of gravity, $m\ s^{-2}$	W	installed power of lamp, $W\ m^{-2}$
H	depth of underground soil for constant temperature, m	w_{ps}	saturated humidity ratio of air at plant temperature, $kg\ kg^{-1}$
h_a	thermal air conductance, $W\ m^{-2}\ K^{-1}$	w_i	humidity ratio of air at indoor temperature, $kg\ kg^{-1}$
h_i, h_o	convection coefficient for indoor and outdoor surfaces, $W\ m^{-2}\ K^{-1}$	Greek letters	
I_b, I_d	direct beam radiation, and diffuse radiation on horizontal surfaces, $W\ m^{-2}$	α_s	factor for estimation of effective solar radiation, dimensionless
I_{bc}, I_{dc}	clear sky direct beam radiation and diffuse radiation, $W\ m^{-2}$	β	angle of inclined surface with horizontal, $^\circ$
I_g	global solar radiation on horizontal surface, $W\ m^{-2}$	γ	surface azimuth angle, $^\circ$
I_{gc}	clear sky global solar radiation on horizontal surface, $W\ m^{-2}$	δ	declination angle of sun, $^\circ$
I_{ex}, I_N, I_{sc}	extraterrestrial solar radiation, sky beam normal radiation, and solar constant ($W\ m^{-2}$)	$\varepsilon_c, \varepsilon_i$	emissivity of cover and indoor components, dimensionless
k_a, k_c, k_s	thermal conductivity of air, cover, and soil, $W\ m^{-1}\ K^{-1}$	$\varepsilon_{sky}, \varepsilon_{clear}$	cloud cover sky emissivity and clear sky emissivity, dimensionless
k	thermal conductivity of ith section in composite wall, $W\ m^{-1}\ K^{-1}$	θ	angle between two radiative surfaces, $^\circ$
L_c, L_f	characteristic length of convective surfaces and plant leaves, m	θ_z	zenith angle of sun, $^\circ$
L_v	latent heat of water vaporization, $J\ kg^{-1}$	θ_i	angle of incidence of surfaces, $^\circ$
MFR	carbon dioxide supply rate in greenhouse, $kg\ m^{-2}\ h^{-1}$	ρ	air density, $kg\ m^{-3}$
M_T	moisture transfer rate, $kg\ s^{-1}$	ρ_r	reflectivity of outdoor ground, dimensionless
n	day of the year, $n = 1$, for January 1st	τ	transmissivity of cover, dimensionless
N	number of air exchange per hour	τ_l	transmissivity of cover to long-wave radiation, dimensionless
N_c	number of layer in covering	μ	dynamic viscosity of air, $kg\ m^{-1}\ s^{-1}$
N_f	number of re-circulation fans	φ	local latitudes, $^\circ$
Nu	Nusselt number, dimensionless	∂	volumetric thermal expansion coefficient, K^{-1}
NHV	net heating value of fuel, $MJ\ kg^{-1}$	σ	Stefan-Boltzmann Constant, $W\ m^{-2}\ K^{-4}$
P	perimeter of greenhouse, m	ω	hour angle, $^\circ$
P_m	motor power rating, W	Δx	thickness of ith section in composite wall, m
P_r	Prandtl number, dimensionless	ΔT	temperature difference, $^\circ C$
PR	CO_2 production rate, kg/kg fuel	Subscripts	
p	atmospheric pressure, kPa	sr, sw	south roof and south wall
p_w	partial pressure of the water vapor, kPa	nr, nw	north roof and north wall
p_{ws}	partial pressure at saturation, kPa	er, ew	east roof and east wall
		wr, ww	west roof and west wall

heating requirement, but the approaches of modeling are different depending on the application of these models. Static models [3,4] were initially developed by considering the heat transfer by conduction, convection, and radiation through the greenhouse cover, but the precision of the static models is very limited ($\pm 25\%$ error) [5]. Reasons include the lack of consideration of the effect of solar radiation in the model, and the prediction is often based on yearly or monthly average meteorological data. Improved static models [6–11] have been developed to increase the accuracy of the simple static models, however, they are valid only for certain types of greenhouses and requires the input of hourly solar radiation data. These intermediate models neglect the heat addition from environmental control systems such as supplemental lighting, CO₂ generators, and various electric motors that contribute significantly to heating in commercial greenhouses. It was estimated the supplemental lighting with high-pressure sodium (HPS) lamps contributed about 25–41% of the total heating requirement of a double-poly greenhouse located in Quebec City [12]. Conversely, complex dynamic models [13–21] have been developed by forming individual energy balance equations for greenhouse interactive components including cover, plant, floor, and indoor air. These models show good precision for predicting the temperature variation of different interactive components in greenhouses, but require complicated modifications to simulate greenhouse heating with a different configuration. For example, Vanthoor et al. [15] developed a dynamic model based on solving the individual energy balance equation of the indoor air, cover, soil, plant, and thermal screen, however, these energy balance equations need complicated modifications for a greenhouse without a thermal screen, or with multilayer of glazing, and that had nontransparent wall. Also, these complex dynamic models neglect the heating contribution of different environmental control systems, and require hourly solar radiation data that are not easily available in some locations.

Energy simulation software including EnergyPlus, EE4, and TRNSYS is available for simulation of energy requirements in different types of buildings. These tools have limitations such as not accurately accounting for the dynamic heat and mass transfer process caused by plant evapotranspiration and other environmental control systems used in greenhouses. Building simulation tools need to be modified to account for heat and mass transfer in a greenhouse, and often need additional models to increase the prediction accuracy [22]. As such, these building simulation tools have limitations for simulation of greenhouse thermal environments, and existing greenhouse lumped estimation models have limitations to consider all of the heat transfer parameters involved in greenhouses. This study develops a time-dependent quasi-steady state thermal model named GREENHEAT to simulate the heating requirement in conventional greenhouses in cold regions.

2. Principles of modelling

The GREENHEAT model was designed to predict the heating requirements of conventional greenhouses under heating

mode when greenhouse temperature and relative humidity (RH) are controlled at a set-point. This model did not consider the cooling mode of the greenhouse, which allows the temperature to rise above the set-point and where RH may be higher or lower than the set-point, because greenhouses in cold regions mostly control indoor temperature through natural ventilation. A greenhouse is a complex physical and biological system including the dynamic heat and mass transfer process and the process of plant photosynthesis. Therefore, several assumptions were considered when developing the model for simulation of heating requirements in conventional greenhouses. The major assumptions for simplification of the model are as follows:

- (1) Greenhouse air is considered to be well-mixed, which means that there is no spatial difference in air temperature because the mechanical air-circulation system is used to avoid air stratification in modern commercial greenhouses.
- (2) Radiative heat exchange between walls and roofs is assumed to be negligible because the temperature is not significantly different when indoor temperature is controlled at the set-point and the air is well circulated.
- (3) The fluctuation of indoor relative humidity is considered to be negligible since the modern commercial greenhouses control relative humidity at an optimum set-point during heating mode.
- (4) The ground heat loss from floor to the depth-ground is considered in a steady state mode since the fluctuation of underground soil temperature is negligible after a certain depth.

The model was developed based on the lumped-system analysis of heat sources and sinks in greenhouses because the indoor temperature and relative humidity in modern commercial greenhouses are usually well controlled and uniform during heating mode. The heat sources in a conventional commercial greenhouse include solar radiation and heat from environmental control systems including supplemental lighting, CO₂ generators, and recirculating fans. The possible heat sinks in greenhouses include heat loss by conduction and convection, heat loss with air exchange, floor and perimeter heat loss, long-wave radiation heat loss, and heat used in the process of plant evapotranspiration.

The general form of heat balance for conventional greenhouses can be given by:

$$Q_h = \text{Sources} - \text{Sinks} \\ = (Q_s + Q_{sl} + Q_{CO_2} + Q_m) - (Q_t + Q_i + Q_g + Q_p + Q_r + Q_e) \quad (1)$$

where Q_h is the heating requirement; Q_s is the net heat gain from solar radiation; Q_{sl} , Q_{CO_2} , and Q_m are the heat gain from supplemental lighting, CO₂ generators, and motors, respectively; Q_t is the heat transfer by conduction and convection; Q_i is the heat transfer caused with air exchange; Q_g is the heat transfer through the greenhouse floor; Q_p is the heat transfer along the perimeter; Q_r is the heat loss by transfer of long-wave radiation; and Q_e is the heat used in the process of plant evapotranspiration.

A negative value (Q_h) from simulation would indicate supplemental heating is needed in the greenhouse, while a positive value indicates cooling needs. A detailed description of the modeling principle for estimation of heat sources and sinks in conventional greenhouses is given in the following sections.

2.1. Heat gain from solar radiation

The solar radiation that passes through the greenhouse cover can be absorbed by greenhouse indoor components; the remaining portion is lost outside of the greenhouse. It is very complicated to estimate the amount of solar insolation effectively used in a greenhouse. As a rough rule, one-half of the available insolation in the greenhouse is converted immediately to sensible heat added to the air, one-quarter is added as latent heat to the greenhouse air, and the last quarter is usually lost [23,24]. The net heat gain from solar radiation can be estimated as follows:

$$Q_s = \alpha_s S \quad (2)$$

The total solar radiation available in the greenhouse depends on several factors, mainly on the transmissivity and orientation of the transparent cover surface. The transparent cover surface area available to receive solar radiation is different depending on the type of greenhouse. Therefore, the solar radiation through different surfaces is estimated separately based on the orientation of the surfaces. Finally, the total solar radiation in greenhouses is the sum of the incoming solar radiation through different surfaces exposed to the outside and can be given by [18]:

$$S = \sum \tau_i A_i I_i \quad (3)$$

where τ_i , and A_i , are the solar transmissivity and area of i th section; and I_i is the total solar radiation available on i th section.

$$\begin{aligned} \sum \tau_i A_i I_i = & \tau_{sr} A_{sr} I_{sr} + \tau_{sw} A_{sw} I_{sw} + \tau_{nr} A_{nr} I_{nr} + \tau_{nw} A_{nw} I_{nw} \\ & + \tau_{ew} A_{ew} I_{ew} + \tau_{ww} A_{ww} I_{ww} + \tau_{er} A_{er} I_{er} \\ & + \tau_{wr} A_{wr} I_{wr} \end{aligned} \quad (3a)$$

2.1.1. Hourly solar radiation through the inclined surface

The solar radiation on any inclined surface includes the beam component from direct beam radiation and the diffuse radiation. The diffuse component composed of isotropic diffuse component received uniformly from the skydome, circum-solar diffuse component resulting from forward scattering of solar radiation and concentrated in an area close to the sun, horizon brightening component concentrated in a band near the horizon and most pronounced in clear skies, and a reflected component that quantifies the radiation reflected from the ground to the tilted surface. Different models (Isotropic sky model, Klucher model, Hay and Davies model, Reindl model, Muneer model, and Perez model) are available for estimating solar radiation on an inclined surface. The isotropic sky model is the simplest model for predicting the solar radiation on the inclined surface with reasonable accuracy as compared to the other more complicated models [25]. According to Liu and Jordan [26], the total solar radiation on an

inclined surface with an angle (β) is given by the isotropic diffuse model:

$$I_i = I_b \frac{\cos \theta_i}{\cos \theta_z} + I_d \left(\frac{1 + \cos \beta}{2} \right) + (I_b + I_d) \rho_r \left(\frac{1 - \cos \beta}{2} \right) \quad (4)$$

The angle of incidence of beam radiation on a surface and the zenith angle of the sun can be expressed as follows [27]:

$$\begin{aligned} \cos \theta_i = & \sin \delta \sin \varphi \cos \beta - \sin \delta \cos \varphi \sin \beta \cos \gamma \\ & + \cos \delta \cos \varphi \cos \beta \cos \omega \\ & + \cos \delta \sin \varphi \sin \beta \cos \gamma \cos \omega + \cos \delta \sin \beta \sin \gamma \sin \omega \end{aligned} \quad (5)$$

$$\cos \theta_z = \cos \varphi \cos \delta \cos \omega + \sin \varphi \sin \delta \quad (6)$$

The declination angle (δ) depends on the n th day of the year, and the hour angle (ω) depends on the time of the day; the required values of these parameters have been calculated according to the ASHRAE Fundamental [28].

2.1.2. Cloud cover solar radiation

The solar radiation data are very important for the analysis of solar application systems and building thermal performance. However, the continuous time series solar radiation data are mainly available in developed countries, whereas very limited resources in the developing countries. Also, the measured data available from a solar station become less accurate for a distance of 50 km from the measuring station [29]. Therefore, the estimation of solar radiation by using cloud cover solar radiation models are convenient because the cloud cover data can be easily obtained from meteorological stations. The solar radiation transmitted through the clouds depends on a number of factors such as the degree of cloudiness, cloud thickness (or height), water content per unit of volume, and the distribution of clouds in relation to the position of the sun in the sky. Cloud cover can significantly affect the availability of solar radiation; about 80–90% of solar radiation intensity can be reduced by thick, low-level, and layered cloud [30]. A few empirical models [31–33] have been developed to estimate solar radiation based on cloud cover information. Previous studies [29,34] have reported that the Kasten-Czeplak model is relatively simple and accurate compared to other cloud cover models. These studies also indicate the Kasten-Czeplak model performs significantly better at estimating global solar radiation with local coefficients (A , B). Therefore, the global solar radiation, diffuse radiation, and direct beam radiation on a horizontal surface under cloud cover can be given as follows [31]:

$$I_g = I_{gc} \left(1 - A \left(\frac{CF}{8} \right)^B \right) \quad (7)$$

$$I_d = I_g \left(0.3 + 0.7 \left(\frac{CF}{8} \right)^2 \right) \quad (8)$$

$$I_b = (I_g - I_d) \quad (9)$$

Clear sky global solar radiation on a horizontal surface (I_{gc}) is the sum of the clear sky direct beam radiation (I_{bc}) and the diffuse radiation component (I_{dc}). Several solar radiation models [28,35–37] have been used for estimating clear sky direct and diffuse radiation on a horizontal surface. The solar

radiation model using the turbidity factor is comparatively simple and convenient for estimating clear sky solar radiation. The clear sky beam and diffuse radiation on any horizontal surfaces and the direct normal irradiance on any terrestrial region can be calculated using the turbidity factor [37]:

$$I_{bc} = I_N \cos \theta_z \quad (10)$$

$$I_{dc} = \frac{1}{3} (I_{ex} - I_N) \cos \theta_z \quad (11)$$

$$I_N = I_{ex} \exp \left[\frac{-T_R}{(0.9 + 9.4 \sin \alpha)} \right], \alpha = 90 - \theta_z \quad (12)$$

According to Spencer [38], the extraterrestrial heat from the sun on a surface normal to the sun's ray on the n th day of the year is given by:

$$I_{ex} = I_{sc} \left[1 + 0.033 \times \cos \left(360 \times \frac{n-3}{365} \right) \right] \quad (13)$$

2.2. Heat gain from environmental control systems

Greenhouses can receive a significant amount of heat from environmental control systems such as supplemental lighting, air circulation fans, and carbon dioxide enrichment equipment. Depending on the types of lamps, the supplemental lighting can return about 75–100% of thermal energy as heat to the greenhouses [39]. The heat addition from supplemental lighting can be estimated by the following equation [28]:

$$Q_{sl} = W F_{hc} F_a A_f \quad (14)$$

Recirculation of indoor air is very important for reducing air stratification in greenhouses because air stagnation may cause abnormal growth of plants and also increase heating requirements. Therefore, recirculation fans are needed in greenhouses in the direction of plant rows at distances not less than 30 times of the diameter of the fan [40]. The instantaneous heat gain from the motors of recirculating fans can be estimated by the following equation [28]:

$$Q_m = N_f \frac{P_m}{E_m} F_{um} F_{lm} \quad (15)$$

The heat gains from CO₂ generators depend on the types of equipment used for carbon dioxide enrichment in greenhouses. A significant amount of heat can be added to a greenhouse when CO₂ is produced from the combustion of fuel in greenhouses. The amount of heat from the CO₂ generators can be estimated by the following equation [28]:

$$Q_{co_2} = 0.278 \times NHV \times MFR \times \frac{A_f}{PR} \quad (16)$$

2.3. Heat transfer by conduction and convection

Conduction and convection are two of the major heat transfer phenomena in greenhouses. Heat transfer through the processes of conduction and convection in transparent and non-transparent envelopes can be calculated as follows:

$$Q_t = (U_t A_t + U_n A_n) \times (T_i - T_o) \quad (17)$$

Depending on the types of greenhouse envelopes, two different approaches have been applied to the estimation of the overall conduction and convection heat transfer coefficient. The overall heat transfer coefficient for transparent surfaces and non-transparent surfaces can be given as [37]:

$$U_t = \left[\frac{1}{h_i} + N_c \times \frac{l_c}{k_c} + (N_c - 1) \times \frac{1}{h_a} + \frac{1}{h_o} \right]^{-1} \quad (18)$$

$$U_n = \left[\frac{1}{h_i} + \sum \frac{k}{\Delta x} + \frac{1}{h_o} \right]^{-1} \quad (19)$$

The estimation of thermal air conductance of air spaces (h_a) for a double-layer greenhouse cover is very complicated. The thermal resistance of air spaces depends on several factors including temperature difference, the direction of heat flow, orientation, the thickness of air spaces, and the emissivity coefficient of the surface that encloses the air spaces [28]. The air gap of greenhouse covers varies between 8 and 12 mm depending on the type of cover; and the emissivity of covers range from 0.2 to 0.9 [6]. Depending on the temperature difference and orientation, the thermal resistance of a 13 mm air spaces varies from 0.45 to 0.13 for an emittance range of 0.2–0.82 [28]. Garzoli and Blackwell [41] determine the thermal resistance of the air spaces for double-layer polyethylene-covered greenhouses by analyzing the rate of heat loss through the cover under different climatic conditions. A value of 3.85 (W m⁻² K⁻¹) for the thermal air conductance of the air spaces (h_a) has been found to give very good agreement between the predicted and the measured rate of heat loss. Therefore, the thermal air conductance of the air spaces is assumed to be 3.85 (W m⁻² K⁻¹) for calculating the overall heat transfer coefficient of a multi-layer greenhouse cover.

Several empirical equations [21,42–45] have been developed to determine the convection coefficient for greenhouse covering surfaces, but these equations are only valid for certain conditions. Therefore, the following equation derived from the Nusselt number can be used to estimate the convective heat transfer coefficient for different surfaces [37]:

$$h = \frac{N_u k_a}{L_c} \quad (20)$$

Indoor air velocity in greenhouses is usually very low compared to the outside, so the heat convection between the cover and the indoor components mainly occurs due to the thermal difference between the surfaces. Therefore, the convection heat transfer at the cover inside surface is considered to be a free convection. The indoor convection coefficient of the transparent surfaces under turbulent flow conditions can be estimated using following relation [37]:

$$h_i = \left(\frac{k_a}{L_c} \right) 0.1 (G_r P_r)^{0.33} (10^9 < G_r P_r < 10^{13}) \quad (21)$$

$$G_r = \frac{g \theta L_c^3 \Delta T}{\mu^2} \quad (22)$$

$$P_r = \frac{\mu C_{pa}}{k_a} \quad (23)$$

For nontransparent vertical surfaces, the indoor convection coefficient is estimated by the following equation [37]:

$$h_i = \left(\frac{k_a}{L_c} \right) 0.1(10^{11})^{0.33} \quad (24)$$

On the other hand, the heat convection between the greenhouse enclosed surface and the outside air is considered to be forced convection; because outdoor airspeed is the dominant force for heat convection from an outside surface. For forced convection, the outdoor convection coefficient under turbulent flow along a plane surface can be expressed as follows [46]:

$$h_o = \left(\frac{k_a}{L_c} \right) 0.037(R_e^{0.8} P_r^{0.33})(R_e > 5 \times 10^5) \quad (25)$$

$$R_e = \frac{\rho V_o L_c}{\mu} \quad (26)$$

2.4. Heat transfer by air exchange

The air exchange in greenhouses transfers a large amount of heat to the outside during the winter season. Depending on air-tightness, wind speed, and inside-outside temperature differences, infiltration represents about 20% of the total heat loss from greenhouses [8]. However, the air exchange caused by ventilation was not considered in the model because the forced ventilation system is shut down during the winter season to minimize heat loss when heating is needed. A natural ventilation system usually operates in greenhouses when the indoor temperature is greater than the set-point temperature and, cooling is needed, but natural ventilation of cooling is not considered in this study since the objective is to estimate heating needs. Thus, the heat loss from greenhouses caused by infiltration can be calculated from the following relation [47]:

$$Q_i = 0.33NV(T_i - T_o) \quad (27)$$

2.5. Heat transfer through floor and perimeter

A large amount of heat can be lost through the greenhouse floor including conduction through the ground and heat transfer along the perimeter. The floor surface temperature is considered to be the same as the indoor temperature. Also, the fluctuation of underground soil temperature over the year at a certain depth becomes negligible, so the steady-state method is applied to estimate heat conduction through the floor [48]. Based on steady-state, one-dimensional heat conduction, the heat transfer through the floor can be estimated by the following equation [11]:

$$Q_s = \frac{k_s}{H} A_f(T_i - T_s) \quad (28)$$

The perimeter heat loss can be calculated using the given equation [28]:

$$Q_p = F_p P(T_i - T_o) \quad (29)$$

2.6. Long-wave radiation transfer from greenhouses

The exchange of long-wave radiation through the transparent cover is responsible for a significant amount of heat loss in greenhouses. The thermal radiation emitted from indoor greenhouse components can be transmitted to the sky or

absorbed in the cover, and part of the radiation can reflect back to the greenhouses. The use of a thermal curtain at night reflects the radiative heat back to the greenhouse. Therefore, it is assumed long-wave radiation loss through a transparent roof would be negligible when the reflective thermal curtain is in operation. This transmitted radiative heat loss from a greenhouse can be expressed as follows [6]:

$$Q_r = [\sigma \epsilon_c A_c F_c (T_i^4 - T_c^4) + \sigma \epsilon_i \tau_i A_f F_{sk} (T_i^4 - T_{sk}^4)] \quad (30)$$

The view factor between the greenhouse and the sky (F_{sk}) is considered to be 1.0 because the greenhouse is enclosed by the sky, which can be considered a black hemisphere [49]. The view factor between the ground and the cover (F_c) can be estimated as follows [50]:

$$F_c = \frac{1 + \cos \theta}{2} \quad (31)$$

The cover temperature is typically a linear function of the indoor and the outdoor temperature [6]. The following linear function can be used to solve cover temperature [51]:

$$T_c = \frac{2}{3} T_o + \frac{1}{3} T_i \quad (32)$$

The effective sky temperature is usually estimated by using the different empirical equations based on the measured ground meteorological parameters including air temperature, air-water vapor tension, and sky cloud cover. However, most of the existing greenhouse thermal models used the simplest models [52,53] based on the outside air temperature without taking into account cloud cover. The estimation of sky temperature taking cloud cover into consideration is more accurate than simple correlations based on air temperature. According to Clark and Allen [54] the effective sky temperature can be estimated as follows:

$$T_{sk} = T_o \epsilon_{sk}^{0.25} \quad (33)$$

$$\epsilon_{sk} = (1 + 0.0224 CF - 0.0035 CF^2 + 0.00028 CF^3) \epsilon_{clear} \quad (34)$$

$$\epsilon_{clear} = 0.787 + 0.7641 \ln \left(\frac{T_{dp}}{273} \right) \quad (35)$$

The dew point temperature of the outside air (T_{dp}) based on the outdoor relative humidity (RH_o) and the dry bulb temperature (T_o) can be calculated by the following equation [55]:

$$T_{dp} = \left(\frac{RH_o}{100} \right)^{0.125} (112 + 0.9 \times T_o) + 0.1 \times T_o - 112 \quad (36)$$

2.7. Heat used in evapotranspiration

Evapotranspiration represents evaporation from the floor or growth media and transpiration from the plants, and it is responsible for a significant amount heat loss in greenhouses. The evaporation from the floor is very complicated to model, and this study considers it to be included in plant evapotranspiration. Evapotranspiration from plant leaves is an evaporative cooling process that reduces the canopy surface temperature. Therefore, greenhouse plants get heat from the solar radiation/indoor air to recover the reduced temperature of the leaf surface. The heat used by plants in the process of plant evapotranspiration can be estimated as follows [56]:

$$Q_e = M_T L_v \quad (37)$$

$$M_T = A_p \rho \frac{w_{ps} - w_i}{(R_a + R_s)} \quad (38)$$

The plant surface area can be calculated from the leaf area index of plants. The saturated humidity ratio at plant temperature and the humidity ratio at indoor air temperature can be calculated as follows [23]:

$$w_{ps} = 0.6219 \frac{P_{ws}}{(P - P_{ws})} \quad (39)$$

$$w_i = 0.6219 \frac{P_w}{(P - P_w)} \quad (40)$$

Previous studies [16,57] show that a negligible temperature difference exists between the plant and the indoor air. Therefore, the water vapor saturation pressure at plant temperature is calculated by assuming plant temperature to be the same as the indoor temperature. The actual vapor pressure can be calculated from the relative humidity and the saturated vapor pressure.

The aerodynamic resistance and stomatal resistance in plant evapotranspiration can be calculated from the following relationships [58,59]:

$$R_a = 220 \times \frac{L_f^{0.2}}{v_i^{0.8}} \quad (41)$$

$$R_s = 200 \left(1 + \frac{1}{\exp(0.05(\tau I_g - 50))} \right) \quad (42)$$

3. Program design

The programming flow chart for simulating the greenhouse heating requirement for the model (GREENHEAT) developed in this study is shown in Fig. 1. The programming language MATLAB was used to develop this computer program for sim-

ulation of heating requirements of conventional greenhouses. The input parameters for simulation include latitude and longitude of the study area, the indoor set-point temperature, relative humidity, air exchange rate, physical and thermal properties of crops and construction materials, and hourly weather data including temperature, relative humidity, cloud cover, and wind speed.

4. Description of the model commercial greenhouse

A four-span gable roof greenhouse with east-west orientation (1125 m²) located near the city of Saskatoon (52.09°N latitude, 106.82°W longitude, and 428 m elevation), covered with air-inflated, double-layer polyethylene film was selected for this study. The north wall of the greenhouse was non-transparent and insulated with extruded polystyrene board and plywood sheathing. A similar non-transparent composite wall was constructed 2.43 m from the ground on the east and west section of the sidewall, and 1.65 m on the south section of wall; twin-wall polycarbonate panels (8 mm) were used for the remaining portion of the sidewall. The hourly weather data (temperature, relative humidity, cloud cover, and wind speed) in Saskatoon for 2014 as collected by the National Solar Radiation Database (NSRDB) were used for simulation of the heating requirements in the study greenhouse. Tomato plants were grown in the greenhouse, so the indoor set-point temperatures were maintained at 21 °C during the daytime and 18 °C during the night. The relative humidity in the study greenhouse was controlled at 75% using the mechanical dehumidifier. Depending on the types of crops, the supplemental lighting is usually turned on when the greenhouse solar radiation reaches below 240–300 W/m² [60], and a photoperiod about 14-h was found to be best for optimum growth of tomatoes in greenhouses [61]. Therefore, it was considered that the supplemental lighting would be turned on when

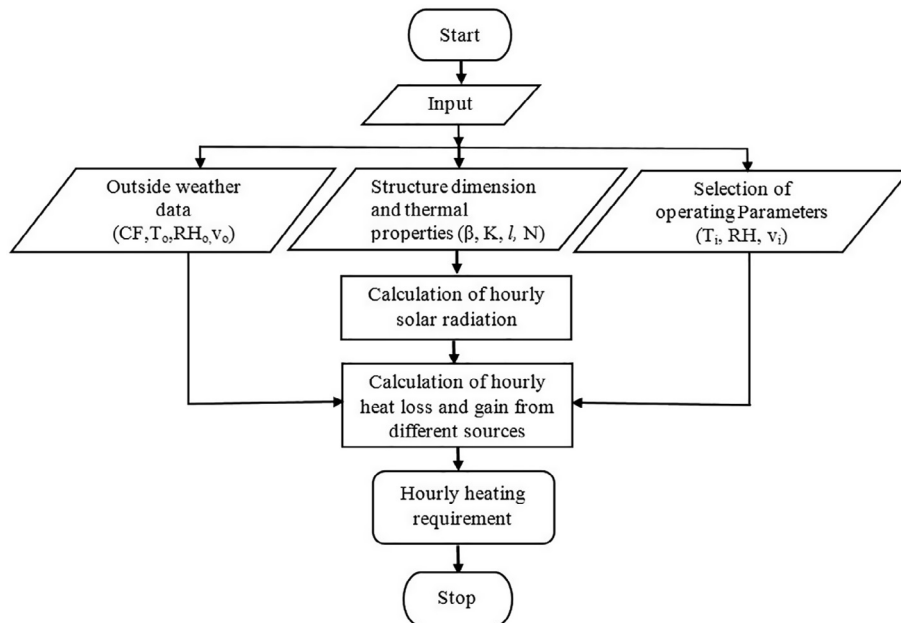


Fig. 1 – Flowchart of the computer program GREENHEAT for simulation of the greenhouse heating requirement.

Table 1 – Constant values of different parameters used in the simulation.

Parameters	Value
<i>Greenhouse characteristics</i>	
Infiltration rate per hour (N) [40]	1.0
Thermal conductivity of plywood (19 mm)	0.12 ($\text{W m}^{-1} \text{K}^{-1}$)
Thermal conductivity of polystyrene insulation (65 mm)	0.03 ($\text{W m}^{-1} \text{K}^{-1}$)
Perimeter heat loss factor (F_p) [28]	0.85 ($\text{W m}^{-1} \text{K}^{-1}$)
Angle of roof (β)	26°
<i>Soil characteristics</i>	
Thermal conductivity of soil (k_s) [28]	1.4 ($\text{W m}^{-1} \text{K}^{-1}$)
Soil temperature (T_s) [48]	15 (°C)
<i>Plant characteristics</i> [62]	
Average leaf area index of tomato (LAI)	2.0
Characteristic length of tomato leaf (L_f)	0.027 (m)
Emissivity coefficient of plants (ϵ_i)	0.9
<i>Covering characteristics</i> [63]	
<i>Poly cover (6 mils)</i>	
Emissivity of IR barrier poly cover (ϵ_c)	0.2
Transmissivity to solar radiation (τ)	0.75
Transmissivity to long-wave radiation (τ_l)	0.29
Thermal conductivity (k_c)	0.33 ($\text{W m}^{-1} \text{K}^{-1}$)
<i>Polycarbonate (8 mm twin-wall)</i>	
Emissivity of polycarbonate (ϵ_c)	0.65
Thermal conductivity (k_c)	0.2 ($\text{W m}^{-1} \text{K}^{-1}$)
Transmissivity to solar radiation (τ)	0.78
Transmissivity coefficient to long-wave radiation (τ_l)	0.03
<i>Other parameters</i>	
Indoor air velocity [39]	0.2 (m/s)
Latent heat of water vaporization (L_v) [28]	2450 (kJ kg^{-1})
Installed lighting wattage (W) [60]	100 (W m^{-2})
Heat conversion factor (F_{hc}) [39]	0.75
Lighting allowance factor (F_a) [28]	1.2
Number of recirculating fans (N_f)	8
Rated power of motors (P_m)	375 (W)
Motor efficiency (E_m)	0.9
Motor load factor (F_{um}) [28]	1.0
Motor use factor (F_{ul}) [28]	1.0
Net heating value of fuel (NHV) [28]	38 (MJ m^{-3} of gas)
Rate of CO ₂ supply in greenhouse (MFR) [39]	4.5 ($\text{g m}^{-2} \text{h}^{-1}$)
CO ₂ production rate (PR) [64]	2.7 (kg kg^{-1} of fuel)
Stefan-Boltzmann Constant (σ)	5.67×10^{-8} ($\text{W m}^{-2} \text{K}^{-4}$)
Reflectivity of outdoor ground (ρ_r)	0.5

solar radiation in the greenhouse was reduced to 250 W m^{-2} , and artificial lighting was turned off between 10 PM and 7 AM to maintain the optimal 14-h photoperiod. The CO₂ generator was considered to be in operation only during the daytime, and a thermal curtain was used at night. Other constant values used in the simulation are listed in Table 1.

5. Results and discussion

5.1. Evaluation of solar radiation sub-model

The model estimated the hourly solar radiation based on the input of hourly cloud cover data from the NSRDB. Fig. 2 shows the scatter plot of the estimated hourly solar radiation from the model versus the data from the NSRDB. The statistical indices including the coefficient of determination (R^2) and

the root mean square error (RMSE) were used to evaluate the adequacy of the solar radiation sub-model. The R^2 and RMSE values are 0.78 and 112.61 W m^{-2} , respectively, which is very close to the results obtained in other studies [29,34,65] to estimate global solar radiation. Based on the statistical indicators, it can be concluded that the estimation of hourly solar radiation from the solar radiation sub-model is reasonably fit with the data from the NSRDB.

5.2. Simulation of the model greenhouse

Fig. 3 shows the annual variation of the simulated heating requirements in the study greenhouse and the monthly average temperature in Saskatoon (52.09°N, −106.82°W) in 2014. The annual total estimated heating requirement in the study greenhouse was about 1380 MJ m^{-2} . The peak heating

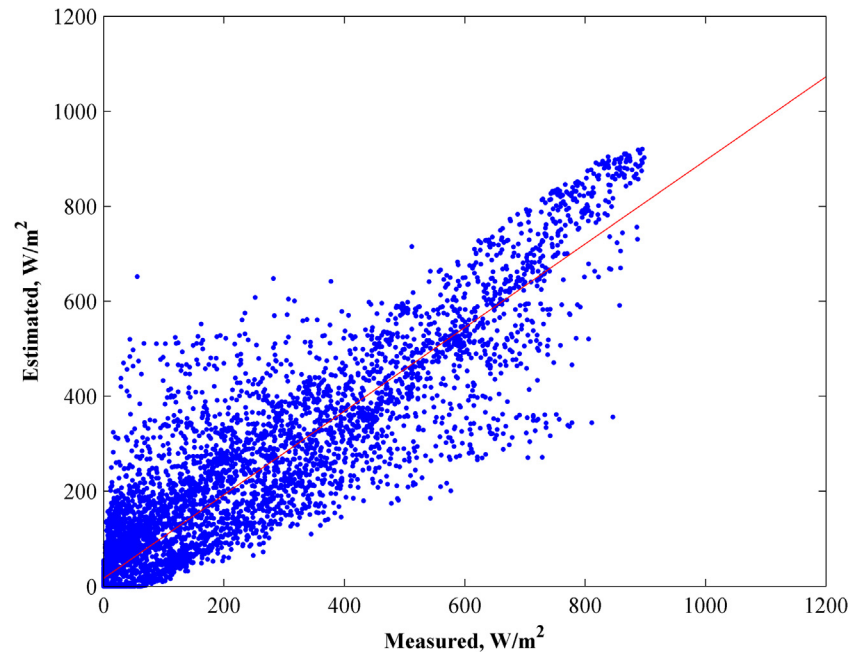


Fig. 2 – The scatter plot of estimated and measured value of global solar radiation on the horizontal surface in Saskatoon.

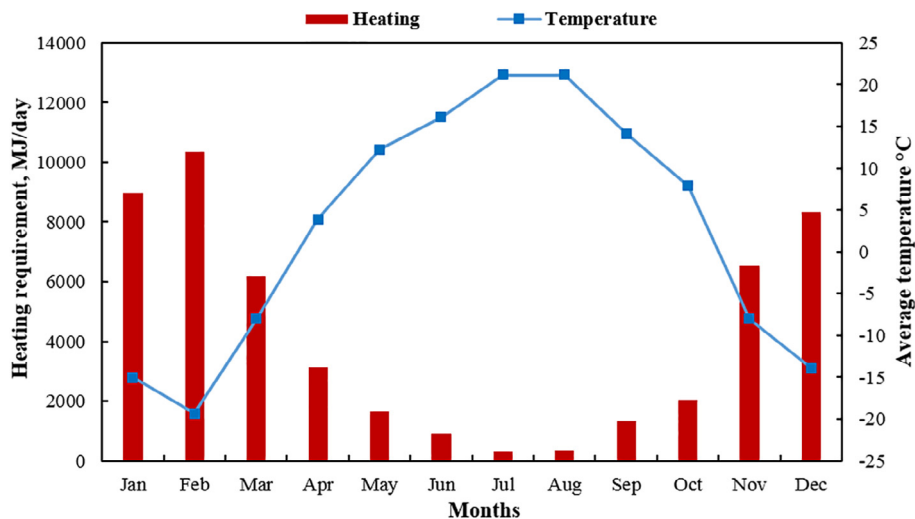


Fig. 3 – Annual variation of the heating requirements in the study greenhouse, and the outside average temperature in Saskatoon.

requirement occurred in February, and was 15.6% higher than the value in January (the lowest average temperature was recorded in February at -19.2°C). Fig. 3 shows that the average temperature in March and November was -8.0°C and -8.03°C , respectively, but the average daily heating requirement in November was about 5.8% higher than the heating requirement in March because the average daily solar heat gain in March was about three times higher than the solar heat gain in November.

5.2.1. Analysis of heat transfers in the greenhouse

Greenhouse heat gain includes the heat from solar radiation and the heat from environmental control systems, including supplemental lighting, CO₂ supply system, and air circulation

system. Fig. 4 shows the variation of heating contributions from solar radiation compared to the total heat gain in the greenhouse. Based on the simulation results, solar radiation contributed about 44–65% of the total heat gain in the greenhouse during the three coldest months (January, February, and December), whereas about 83–86% during the summer months (June–August).

The heating contribution of different environmental control systems is shown in Fig. 5. The predicted heat gains from environmental control systems were about 13–56% of the total greenhouse heat gain over the year. Fig. 5 indicates the heat gain from supplemental lighting decreased with the increase of day length because the artificial photoperiod decreased with the increase of day length in summer; an

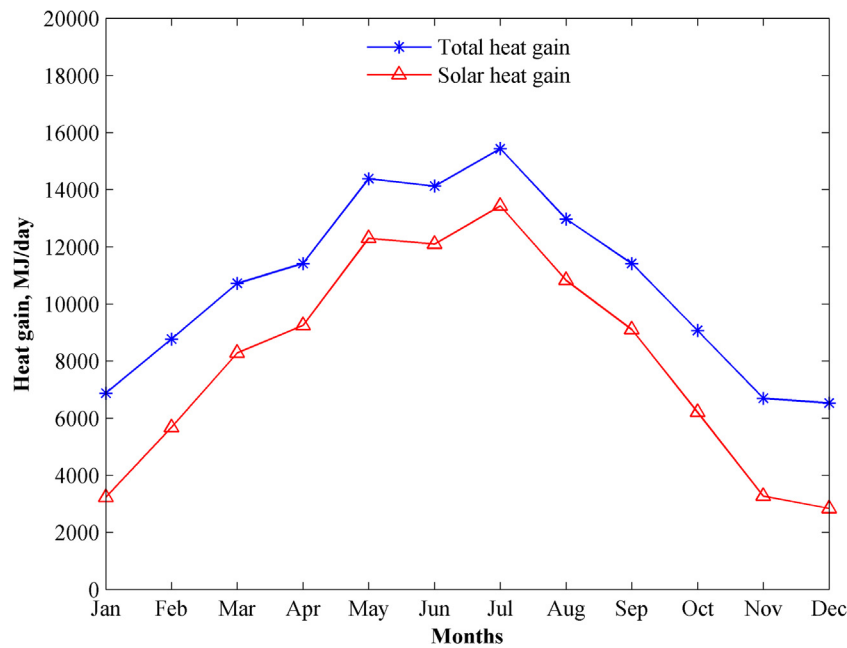


Fig. 4 – Annual variation of heating contribution from solar radiation compared to the total heat gain.

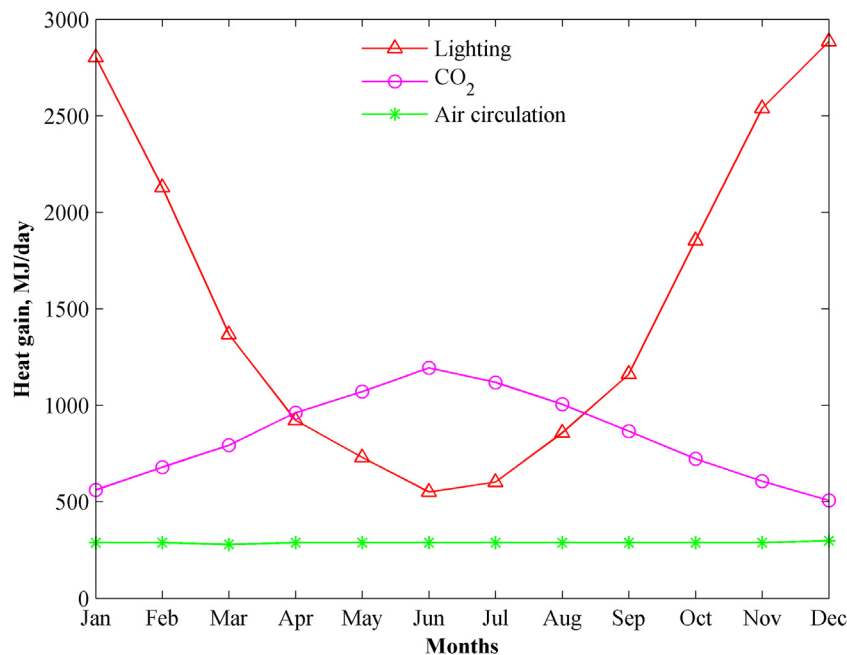


Fig. 5 – Annual variation of the heat gains from different environmental control systems.

opposite trend was true for the CO₂ supply system because the operating time of CO₂ generator increased with an increase in day length. However, the heat gain from electric motors was constant for the entire simulation period since the recirculating fans were considered to be in operation all the time. The heat contribution from supplemental lighting mainly depends on the installed wattage of the lighting system and the period of artificial lighting. At winter solstice in 2014, the HSP lamp with an installed capacity of 100 W/m² contributed about 38.0% of the total heating requirement in the study greenhouse with eight hours of artificial lighting.

Brault et al. [12] reported similar heating contributions from HPS lamps in a double-polyethylene greenhouse in Quebec, Canada. The CO₂ generators contributed about 6.5% of the total heating requirement when operating for seven hours during the daytime, whereas 24-hour operation of recirculating fans contributed about 3.8% of the total heating demand. The predicted result indicates environmental control systems can supply a significant amount of heat to a greenhouse, which can significantly reduce heating demands. However, most of the previous models ignore the heat gain from these environmental control systems, so that a significant error

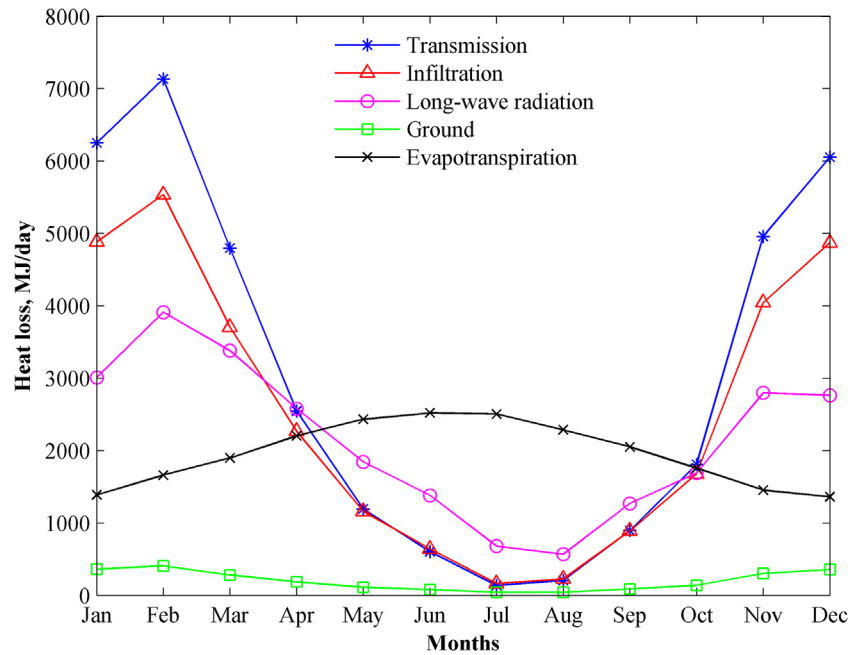


Fig. 6 – Annual variation of heat loss by different heat transfer mechanisms from the study greenhouse during heating mode.

arises in simulation when the heat addition from these environmental systems is neglected.

Fig. 6 shows the annual variation of heat loss from the study greenhouse by different heat transfer mechanisms such as transmission heat loss, infiltration loss, long-wave radiation loss, heat loss through the ground, and heat loss for evapotranspiration. The similar trend of heat losses was found in a conventional greenhouse in the UK [66], however, the magnitude of heat loss by different mechanisms would be different depending on the greenhouse envelope and their location. Since February was the coldest month in 2014 (Fig. 3), the amount of heat loss from the greenhouse was highest in February. About 40% of the total heat from the study greenhouse was lost through conduction and convection during the coldest three months of the year. The heat loss caused by infiltration was the second largest source of heat loss (up to 32%) in the greenhouse, followed by the heat loss caused by long-wave radiation (up to 21%). After March, the long-wave radiation loss increased because the operation hour of the energy curtain decreased with an increase in day length. The heat loss caused by the process of evapotranspiration from plants accounted for about 9.0% during extremely cold periods, while the heat used in the process of evapotranspiration was found to be the highest in summer and spring because evapotranspiration depends greatly on solar radiation. The heat loss along the perimeter and the vertical conduction through the soil fluctuated mildly over the year because the underground temperature was considered to be constant over the year. A very small amount of heat (about 3.0%) was lost through the floor of the study greenhouse. The thermal analysis indicates more than 80% of greenhouse heat is lost by transmission, infiltration, and exchange of long-wave radiation. Therefore, the use of insulation and construction materials with a high thermal resistance, infrared radiation (IR) barrier glazing, and reflective thermal screens

at night would significantly reduce greenhouse heating requirements at high northern latitudes. The intensity of heat loss through different mechanisms could be different depending on the type of structure, materials used for construction, management of environmental control systems, and also, importantly, the location of the greenhouse. However, the thermal analysis of different heat sources and sinks in the greenhouse at high northern latitudes would be helpful for greenhouse growers and researchers for analyzing the energy efficient greenhouse production under similar cold climates.

5.3. Validation of the model

The actual heating requirement in the study greenhouse for the year of 2014 from March to October was calculated based on the data collected from the commercial greenhouse. The data for the four coldest months were ignored for validation because the greenhouse started its operation in mid-February and continued until mid-November. The monthly heating requirement was calculated based on the heating value (38 MJ kg^{-1}) and the retail rate ($\text{CAD } 0.286 \text{ m}^{-3}$) of natural gas. The predicted heating requirement was calculated without considering the heating contribution from the supplemental lighting because no supplemental lighting was provided in the greenhouse in 2014. The comparison of the predicted and the actual heating requirement is shown in Fig. 7. Fig. 7 shows that the predicted heating requirement in the greenhouse is close to the actual heating requirement, except for the summer months when the natural ventilation system was in operation very frequently to control the indoor relative humidity instead of the mechanical dehumidifier. Therefore, the air-exchange rate during summer months (July–August) could be relatively higher that draws the cool outside air into the greenhouse, thereby increasing the

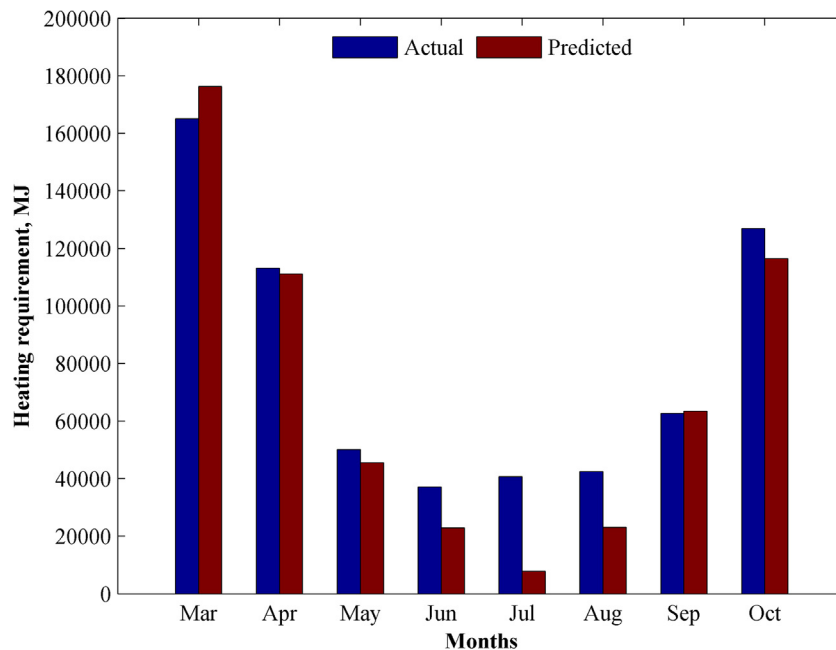


Fig. 7 – Comparison of predicted and actual heating requirements in the study greenhouse.

greenhouse heating requirement. Also, the actual heating requirement during the summer months is very close to the heating requirement in May, but the average temperature during the summer months was relatively high compared to the average temperature in May. Hence, the conclusion regarding the model validation was made without considering the data from June to August. The validation of the model shows that the errors in heating prediction for the winter months is less than 9.0%, and 4.6% is the average simulation error. Previous studies [5,8,57] reported that the average percent error in simulation of about 10% is reasonably acceptable for greenhouse thermal modeling. The percent error of the simple static models [3,4] is about 25%, and close to 11% for the improved static models [67,68], whereas error is less than 10% for the complex dynamic models [14,18,43,69]. Therefore, it can be concluded that GREENHEAT can predict the heating requirement of conventional greenhouses with good accuracy.

6. Conclusions

In this study, a quasi-steady state thermal model named GREENHEAT was developed based on the lumped estimation of heat sources and sinks in greenhouses. The developed model considered most of the heat sources and sinks in a conventional greenhouse, including the hourly variation of solar radiation, heat gain from environmental control systems, and heat used for plant evapotranspiration. The simulated results indicate that the environmental control systems (supplemental lighting, CO₂ generators, and motors of recirculating fans) contribute up to 56% of heating requirements in the study greenhouse. The transmission heat loss by conduction and convection is the largest component of heat loss (40%) during the winter periods followed by the infiltration and long-wave radiation loss, respectively. The validation shows that the average error in prediction is less than 5%,

therefore, the model can be considered sufficiently accurate for simulation of heating requirements in conventional greenhouses. The model will be beneficial for researchers, engineers, and growers to assist with their decision-making regarding energy-efficient designs and feasibility analysis of greenhouse production in cold northern climates.

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